

Solution	Marks	
SECTION A 1. (a) $(1 + ax)^n = 1 + C_1^n(ax) + C_2^n(ax)^2 + \dots$ Coefficient of $x = na = -12 \implies a = -\frac{12}{n}$ Coefficient of $x^2 = \frac{n(n-1)}{2}a^2 = 54$ $\frac{n(n-1)}{2} \left(-\frac{12}{n}\right)^2 = 54$ $\frac{144(n-1)}{2n} = 54 \implies 72(n-1) = 54n \implies 18n = 72 \implies n = 4$ $a = -\frac{12}{4} = -3$ (b) Coefficient of $x^3 = C_3^n a^3 = C_3^4 (-3)^3$ $= 4 \times (-27) = -108$	1M 1M 1A 1A 1M 1A	

Solution	Marks	
2. (a) $(2 + 4x)^n = \sum_{k=0}^n C_k^n 2^{n-k} (4x)^k = \sum_{k=0}^n C_k^n 2^{n+k} x^k$ $u_0 = 2^n, u_1 = C_1^n 2^{n+1} = 2n \cdot 2^n, u_2 = C_2^n 2^{n+2} = 2n(n-1) \cdot 2^n$ Given $2u_1 + 8u_0 = u_2$: $2(2n \cdot 2^n) + 8(2^n) = 2n(n-1) \cdot 2^n$ $4n + 8 = 2n^2 - 2n \implies 2n^2 - 6n - 8 = 0 \implies n^2 - 3n - 4 = 0$ $(n-4)(n+1) = 0$ Since n is a positive integer, $n = 4$. (b) Substitute $n = 4$: $(2 + 4x)^4 = \sum_{k=0}^4 C_k^4 2^{4-k} (4x)^k = 16 + 128x + 384x^2 + 512x^3 + 256x^4$ $\therefore 0 \cdot u_0 + 1 \cdot u_1 + 2 \cdot u_2 + \cdots + n \cdot u_n = 0 \cdot 16 + 1 \cdot 128 + 2 \cdot 384 + 3 \cdot 512 + 4 \cdot 256 = 3456$	1M 1M 1A 1M 1A	

Solution	Marks	
3. (a) $\left(1 + \frac{1}{x}\right)^2 \left(kx + \frac{1}{x}\right)^4 = \left(1 + \frac{2}{x} + \frac{1}{x^2}\right) \sum_{r=0}^4 C_r^4 (kx)^{4-r} \left(\frac{1}{x}\right)^r$ $= \left(1 + \frac{2}{x} + \frac{1}{x^2}\right) \sum_{r=0}^4 C_r^4 k^{4-r} x^{4-2r}$ $= \sum_{r=0}^4 C_r^4 k^{4-r} x^{4-2r} + \sum_{r=0}^4 2C_r^4 k^{4-r} x^{4-2r-1} + \sum_{r=0}^4 C_r^4 k^{4-r} x^{4-2r-2}$ Constant term $= C_2^4 k^{4-2} + C_1^4 k^{4-1} = 6k^2 + 4k^3 = 0$ $2k^2(2 + 3k) = 0 \implies k = 0$ or $k = -\frac{2}{3}$ (b) Coefficient of x^{-n} in $\left(1 + \frac{1}{x}\right)^2 \left(kx + \frac{1}{x}\right)^n$: $\left(1 + \frac{2}{x} + \frac{1}{x^2}\right) \sum_{r=0}^n C_r^n k^{n-r} x^{n-2r}$ To get x^{-n} term: - From $1 \times x^{-n}$: $n - 2r = -n \implies 2r = 2n \implies r = n \implies 1 \cdot C_n^n k^0 = 1$ - From $\frac{2}{x} \times x^{-n+1}$: $n - 2r = -n + 1 \implies 2r = 2n - 1$ (no integer solution) - From $\frac{1}{x^2} \times x^{-n+2}$: $n - 2r = -n + 2 \implies 2r = 2n - 2 \implies r = n - 1 \implies 1 \cdot C_{n-1}^n k^1 = nk$ Total coefficient of $x^{-n} = 1 + nk$ When $k = 0$, $1 + nk = 1 + n(0) = 1$. Thus, the claim that it "must not be 1" is incorrect.	1M 1M 1A 1M 1A	

Solution	Marks	
4. (a) $(au + t)^n = \sum_{r=0}^n C_r^n (au)^r t^{n-r} = \sum_{r=0}^n \alpha_r u^r \implies \alpha_r = C_r^n a^r t^{n-r}$ $\alpha_1 = nat^{n-1} = 12 \dots\dots(1)$ $\alpha_2 = \frac{n(n-1)}{2} a^2 t^{n-2} = 60 \dots\dots(2)$ $\alpha_3 = \frac{n(n-1)(n-2)}{6} a^3 t^{n-3} = 160 \dots\dots (3)$ $\frac{(2)}{(1)} \implies \frac{n-1}{2} \cdot \frac{a}{t} = 5 \implies \frac{a}{t} = \frac{10}{n-1}$ $\frac{(3)}{(2)} \implies \frac{n-2}{3} \cdot \frac{a}{t} = \frac{8}{3} \implies \frac{a}{t} = \frac{8}{n-2}$ $\frac{10}{n-1} = \frac{8}{n-2} \implies 10n - 20 = 8n - 8 \implies 2n = 12 \implies n = 6$ Then $\frac{a}{t} = 2 \implies a = 2t$. Substitute into (1): $6(2t)t^5 = 12 \implies t^6 = 1 \implies t = 1$ (since $t \geq 0$). $a = 2(1) = 2$. Thus, $a = 2, t = 1, n = 6$. (b) $(au + bv)^n = \sum_{r=0}^n C_r^n (au)^{n-r} (bv)^r = \sum_{r=0}^n \beta_r v^r \implies \beta_r = C_r^n (au)^{n-r} b^r$ $\alpha_4 = C_4^6 a^4 t^2 = 15(2)^4 (1)^2 = 240$ $\beta_4 = C_4^6 (au)^2 b^4 = 15(2u)^2 b^4 = 60b^4 u^2$ $\beta_4 - \alpha_4 = 60b^4 u^2 - 240 = 60(b^2 u - 2)(b^2 u + 2)$ $\therefore \beta_4 - \alpha_4$ is divisible by $b^4 + 2$.	1M 1M 1A 1A 1M 1M 1A	

Solution	Marks	
5. (a) $(x^{-2} + x)^5(x^3 - x^{-2})^2 = (x^{-2} + x)^5(x^6 - 2x + x^{-4})$ $(x^{-2} + x)^5 = \sum_{r=0}^5 C_r^5 (x^{-2})^{5-r} x^r = \sum_{r=0}^5 C_r^5 x^{3r-10}$ To find coefficient of x^{-4} in $\sum_{r=0}^5 C_r^5 x^{3r-10} \times (x^6 - 2x + x^{-4})$: - With x^6: $3r - 10 + 6 = -4 \implies 3r = 0 \implies r = 0 \implies C_0^5(1) = 1$ - With $-2x$: $3r - 10 + 1 = -4 \implies 3r = 5$ (no integer solution) - With x^{-4}: $3r - 10 - 4 = -4 \implies 3r = 10$ (no integer solution) Coefficient of $x^{-4} = 1$ (b) To find coefficient of x^2: - With x^6: $3r - 10 + 6 = 2 \implies 3r = 6 \implies r = 2 \implies C_2^5(1) = 10$ - With $-2x$: $3r - 10 + 1 = 2 \implies 3r = 11$ (no integer solution) - With x^{-4}: $3r - 10 - 4 = 2 \implies 3r = 16$ (no integer solution) Coefficient of $x^2 = 10 \neq 0$. Therefore, I disagree with the claim.	1M 1M 1A 1M 1M 1A	

Solution	Marks	
6. (a) $(1 + kx)^n = 1 + C_1^n(kx) + C_2^n(kx)^2 + C_3^n(kx)^3 + \dots$ Coefficient of $x = nk$ Coefficient of $x^2 = \frac{n(n-1)}{2}k^2$ Coefficient of $x^3 = \frac{n(n-1)(n-2)}{6}k^3$ Given $\text{Coeff}(x^2) : \text{Coeff}(x^3) = 1 : 2$: $\frac{\frac{n(n-1)}{2}k^2}{\frac{n(n-1)(n-2)}{6}k^3} = \frac{1}{2} \implies \frac{3}{(n-2)k} = \frac{1}{2} \implies (n-2)k = 6 \implies k = \frac{6}{n-2}$ Given $\text{Coeff}(x) + \text{Coeff}(x^2) = 50$: $nk + \frac{n(n-1)}{2}k^2 = 50$ $n \left(\frac{6}{n-2} \right) + \frac{n(n-1)}{2} \left(\frac{36}{(n-2)^2} \right) = 50$ $\frac{6n(n-2) + 18n(n-1)}{(n-2)^2} = 50 \implies 24n^2 - 30n = 50(n^2 - 4n + 4)$ $26n^2 - 170n + 200 = 0 \implies 13n^2 - 85n + 100 = 0$ $(13n - 20)(n - 5) = 0$ Since n is an integer, $n = 5$. $k = \frac{6}{5-2} = 2. \text{ Thus, } n = 5, k = 2.$ (b) $(x^2 - \frac{1}{2x})^6 P(x) = \sum_{r=0}^6 C_r^6 (x^2)^{6-r} \left(-\frac{1}{2x}\right)^r (1+2x)^5$ $= \sum_{r=0}^6 C_r^6 \left(-\frac{1}{2}\right)^r x^{12-3r} \cdot \sum_{s=0}^5 C_s^5 2^s x^s$ For constant term, exponent of x is $12 - 3r + s = 0 \implies 3r - s = 12$. Since $0 \leq s \leq 5$ and $0 \leq r \leq 6$: - If $r = 4$: $3(4) - s = 12 \implies s = 0$ - If $r = 5$: $3(5) - s = 12 \implies s = 3$ Constant term $= C_4^6 \left(-\frac{1}{2}\right)^4 C_0^5 2^0 + C_5^6 \left(-\frac{1}{2}\right)^5 C_3^5 2^3$ $= 15 \times \frac{1}{16} \times 1 \times 1 + 6 \times \left(-\frac{1}{32}\right) \times 10 \times 8$ $= \frac{15}{16} - \frac{480}{32} = \frac{15}{16} - 15 = -\frac{225}{16}$	1M 1M 1M 1A 1M 1M 1M 1A	

Solution	Marks	
7. (a) (i) $f(x) = (1 + kx)^n$ Coefficient of $x = nk$ Coefficient of $x^2 = \frac{n(n-1)}{2}k^2$ (ii) $\frac{n(n-1)}{2}k^2 = 12(nk)$ Since $n > 2$ and $k > 0$, divide both sides by nk: $\frac{(n-1)k}{2} = 12 \implies k(n-1) = 24$ (b) (i) $g(x) = (kx - 2)(1 + x)^n = (kx - 2) \sum_{r=0}^n C_r^n x^r$ Coefficient of $x = kC_0^n - 2C_1^n = k - 2n$ (ii) Constant term of $g(x) = -2C_0^n = -2$ Given Constant term + Coeff(x) = -6: $-2 + (k - 2n) = -6 \implies k - 2n = -4 \implies k = 2n - 4$ Substitute into $k(n-1) = 24$: $(2n - 4)(n - 1) = 24 \implies 2(n - 2)(n - 1) = 24 \implies n^2 - 3n + 2 = 12$ $n^2 - 3n - 10 = 0 \implies (n - 5)(n + 2) = 0$ Since $n > 2$, $n = 5$. $k = 2(5) - 4 = 6. \text{ Thus, } n = 5, k = 6.$ (c) Coefficient of x^2 in $g(x) = kC_1^n - 2C_2^n$ $= kn - 2 \cdot \frac{n(n-1)}{2} = kn - n(n-1)$ Substitute $n = 5, k = 6$: $\text{Coefficient of } x^2 = 6(5) - 5(4) = 30 - 20 = 10$	1A 1A 1M 1A 1M + 1A 1M 1M 1A 1M 1M 1A	

Solution	Marks	
8. (a) (i) $f(x) = \left(x^2 - \frac{1}{x}\right) \left(ax + \frac{b}{x^2}\right)^n = \left(x^2 - \frac{1}{x}\right) \sum_{r=0}^n C_r^n (ax)^{n-r} \left(\frac{b}{x^2}\right)^r$ $= \left(x^2 - \frac{1}{x}\right) \sum_{r=0}^n C_r^n a^{n-r} b^r x^{n-3r}$ To get x^{n-1} term: $x^2 \times x^{n-3} \implies n - 3r = n - 3 \implies r = 1 \implies C_1^n a^{n-1} b^1 = na^{n-1}b$ $-\frac{1}{x} \times x^n \implies n - 3r = n \implies r = 0 \implies -C_0^n a^n = -a^n$ Coefficient of $x^{n-1} = na^{n-1}b - a^n$ (ii) If $a = nb \implies b = \frac{a}{n}$: To get x^{n-4} term in $f(x)$: - From $x^2 \times x^{n-6}$: $n - 3r = n - 6 \implies r = 2 \implies C_2^n a^{n-2} b^2 = \frac{n(n-1)}{2} a^{n-2} b^2$ - From $-\frac{1}{x} \times x^{n-3}$: $n - 3r = n - 3 \implies r = 1 \implies -C_1^n a^{n-1} b = -na^{n-1}b$ Coeff(x^{n-4}) = $\frac{n(n-1)}{2} a^{n-2} \left(\frac{a}{n}\right)^2 - na^{n-1} \left(\frac{a}{n}\right)$ $= \frac{n-1}{2n} a^n - a^n = \left(\frac{n-1-2n}{2n}\right) a^n = -\frac{n+1}{2n} a^n$ (b) (i) $g(x) = \left(x^2 + \frac{1}{x}\right) \left(ax - \frac{k}{x^2}\right)^{n+1} = \left(x^2 + \frac{1}{x}\right) \sum_{r=0}^{n+1} C_r^{n+1} (ax)^{n+1-r} \left(-\frac{k}{x^2}\right)^r$ To get x^n term: - From $x^2 \times x^{n-2}$: $n + 1 - 3r = n - 2 \implies 3r = 3 \implies r = 1$ $\implies C_1^{n+1} a^n (-k)^1 = -(n+1)a^n k$ - From $\frac{1}{x} \times x^{n+1}$: $n + 1 - 3r = n + 1 \implies r = 0 \implies C_0^{n+1} a^{n+1} = a^{n+1}$ Coeff(x^n) = $a^{n+1} - (n+1)a^n k = 0 \implies k = \frac{a}{n+1}$ (ii) To get x^{n-3} term in $g(x)$: - From $x^2 \times x^{n-5}$: $n + 1 - 3r = n - 5 \implies 3r = 6 \implies r = 2$ $\implies C_2^{n+1} a^{n-1} (-k)^2 = \frac{(n+1)n}{2} a^{n-1} k^2$ - From $\frac{1}{x} \times x^{n-2}$: $n + 1 - 3r = n - 2 \implies 3r = 3 \implies r = 1$ $\implies C_1^{n+1} a^n (-k)^1 = -(n+1)a^n k$ Substitute $k = \frac{a}{n+1}$: Coeff(x^{n-3}) = $\frac{(n+1)n}{2} a^{n-1} \left(\frac{a}{n+1}\right)^2 - (n+1)a^n \left(\frac{a}{n+1}\right) = \frac{n}{2(n+1)} a^{n+1} - a^{n+1}$ $= \left(\frac{n-2n-2}{2(n+1)}\right) a^{n+1} = -\frac{n+2}{2(n+1)} a^{n+1}$ (c) Given $a = 1$:	1M 1A 1M 1A 1M 1A 1M 1A	

Solution	Marks	
Coeff of x^{n-4} in $f(x) = -\frac{n+1}{2n}$ Coeff of x^{n-3} in $g(x) = -\frac{n+2}{2(n+1)}$ Ratio = $\frac{-\frac{n+1}{2n}}{-\frac{n+2}{2(n+1)}} = \frac{(n+1)^2}{n(n+2)} = \frac{25}{24}$ $24(n^2 + 2n + 1) = 25(n^2 + 2n) \implies 24n^2 + 48n + 24 = 25n^2 + 50n$ $n^2 + 2n - 24 = 0 \implies (n + 6)(n - 4) = 0$ Since $n > 3$, $n = 4$. $k = \frac{1}{4+1} = \frac{1}{5}$. Thus, $n = 4, k = \frac{1}{5}$.	1M 1M 1A	

Solution	Marks	
9.		
(a) (i) $1 + F_n(x) = 1 + \sum_{k=1}^n C_k^n \left(\frac{x}{1+2x}\right)^k = \sum_{k=0}^n C_k^n \left(\frac{x}{1+2x}\right)^k \cdot 1^{n-k}$	1M	
$= \left(1 + \frac{x}{1+2x}\right)^n = \left(\frac{1+2x+x}{1+2x}\right)^n = \left(\frac{1+3x}{1+2x}\right)^n$	1A	
(ii)		
$1 + F_n(x) = \left(\frac{1+3x}{1+2x}\right)^n$		
$(1+2x)^n(1+F_n(x)) = (1+3x)^n$	1M	
$(1+2nx+\dots)(1+c_0+c_1x+\dots) = 1+3nx+\dots$		
$1+c_0+(2n+c_1)x+\dots = 1+3nx+\dots$		
$\implies c_1 = n$	1A	
(b) (i) $G_n(x) = (1+2x)^n \cdot [1+F_n(x)] = (1+2x)^n \left(\frac{1+3x}{1+2x}\right)^n = (1+3x)^n$	1A	
(ii) Since $G_n(x) = (1+3x)^n = \sum_{r=0}^n C_r^n (3x)^r = \sum_{r=0}^n C_r^n 3^r x^r$, we have $b_r = C_r^n 3^r$.	1M	
$\frac{b_r}{b_{r-1}} = \frac{C_r^n 3^r}{C_{r-1}^n 3^{r-1}} = 3 \cdot \frac{\frac{n!}{r!(n-r)!}}{\frac{n!}{(r-1)!(n-r+1)!}}$	1M	
$= 3 \cdot \frac{(r-1)!}{r!} \cdot \frac{(n-r+1)!}{(n-r)!} = 3 \cdot \frac{1}{r} \cdot (n-r+1) = \frac{3(n-r+1)}{r}$	1M + 1A	
(c) Given $b_{r-1} : b_r : b_{r+1} = 1 : 3 : 7$:		
$\frac{b_r}{b_{r-1}} = 3 \implies \frac{3(n-r+1)}{r} = 3 \implies n-r+1 = r \implies n = 2r-1$	1M	
$\frac{b_{r+1}}{b_r} = \frac{7}{3} \implies \frac{3(n-(r+1)+1)}{r+1} = \frac{7}{3} \implies \frac{3(n-r)}{r+1} = \frac{7}{3}$		
Substitute $n = 2r-1$:		
$\frac{3(2r-1-r)}{r+1} = \frac{7}{3} \implies \frac{3(r-1)}{r+1} = \frac{7}{3} \implies 9(r-1) = 7(r+1)$	1M	
$9r-9 = 7r+7 \implies 2r = 16 \implies r = 8$		
$n = 2(8) - 1 = 15. \text{ Thus, } n = 15, r = 8.$	1A	

Solution	Marks	
SECTION B		
10.		
(a) $M^2 = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$	1M + 1A	
(b) $X = \sum_{r=0}^5 C_r^5 2^{5-r} M^r = (2I + M)^5$	1M	
Since $M^2 = I$, for any non-negative integer k : $M^{2k} = I$, $M^{2k+1} = M$		
$(2I + M)^5 = \sum_{r=0}^5 C_r^5 2^{5-r} M^r$		
$= (C_0^5 2^5 + C_2^5 2^3 + C_4^5 2^1) I + (C_1^5 2^4 + C_3^5 2^2 + C_5^5 2^0) M$	1M	
$a = C_0^5(32) + C_2^5(8) + C_4^5(2) = 1(32) + 10(8) + 5(2) = 32 + 80 + 10 = 122$	1A	
$b = C_1^5(16) + C_3^5(4) + C_5^5(1) = 5(16) + 10(4) + 1(1) = 80 + 40 + 1 = 121$	1A	
$X = 122I + 121M = 122 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 121 \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$		
$= \begin{pmatrix} 122 + 242 & -121 \\ 363 & 122 - 242 \end{pmatrix} = \begin{pmatrix} 364 & -121 \\ 363 & -120 \end{pmatrix}$	1A	

Solution	Marks	
11. (a) Let $P(n)$ be the proposition "$A^n = \frac{1}{2} \begin{pmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{pmatrix}$". For $n = 1$: R.H.S. $= \frac{1}{2} \begin{pmatrix} 3 - 1 & 3 + 1 \\ 3 + 1 & 3 - 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = A = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ is true for some positive integer k, i.e. $A^k = \frac{1}{2} \begin{pmatrix} 3^k + (-1)^k & 3^k - (-1)^k \\ 3^k - (-1)^k & 3^k + (-1)^k \end{pmatrix}.$ For $n = k + 1$: $\begin{aligned} A^{k+1} &= A^k A = \frac{1}{2} \begin{pmatrix} 3^k + (-1)^k & 3^k - (-1)^k \\ 3^k - (-1)^k & 3^k + (-1)^k \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} [3^k + (-1)^k] + 2[3^k - (-1)^k] & 2[3^k + (-1)^k] + [3^k - (-1)^k] \\ 2[3^k + (-1)^k] + [3^k - (-1)^k] & [3^k + (-1)^k] + 2[3^k - (-1)^k] \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 3 \cdot 3^k - (-1)^k & 3 \cdot 3^k + (-1)^k \\ 3 \cdot 3^k + (-1)^k & 3 \cdot 3^k - (-1)^k \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 3^{k+1} + (-1)^{k+1} & 3^{k+1} - (-1)^{k+1} \\ 3^{k+1} - (-1)^{k+1} & 3^{k+1} + (-1)^{k+1} \end{pmatrix} \end{aligned}$ $\therefore P(k + 1)$ is true. By M.I., $P(n)$ is true for all positive integers n. (b) (i) $A^2 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ $A^2 - 2A - 3I = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} - 2 \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} - 3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (ii) $B_n = \sum_{r=0}^n C_r^n A^{2r} = (I + A^2)^n$ From (b)(i), $A^2 = 2A + 3I \implies I + A^2 = 2A + 4I = 2(A + 2I)$ $B_n = [2(A + 2I)]^n = 2^n (A + 2I)^n$ $(A + 2I)^n = \sum_{k=0}^n C_k^n A^k (2I)^{n-k}$ $(A + 2I)^n = \sum_{k=0}^n C_k^n 2^{n-k} \frac{1}{2} \begin{pmatrix} 3^k + (-1)^k & 3^k - (-1)^k \\ 3^k - (-1)^k & 3^k + (-1)^k \end{pmatrix}$ For the top-left and bottom-right elements:	1M 1M 1M 1A 1M 1A 1M 1M	

Solution	Marks	
$\frac{1}{2} \sum_{k=0}^n C_k^n 2^{n-k} (3^k + (-1)^k) = \frac{1}{2} \sum_{k=0}^n C_k^n 2^{n-k} 3^k + \frac{1}{2} \sum_{k=0}^n C_k^n 2^{n-k} (-1)^k$ $= \frac{1}{2} (2+3)^n + \frac{1}{2} (2+(-1))^n = \frac{1}{2} (5^n + 1)$ For the top-right and bottom-left elements: $\frac{1}{2} \sum_{k=0}^n C_k^n 2^{n-k} (3^k - (-1)^k) = \frac{1}{2} \sum_{k=0}^n C_k^n 2^{n-k} 3^k - \frac{1}{2} \sum_{k=0}^n C_k^n 2^{n-k} (-1)^k$ $= \frac{1}{2} (2+3)^n - \frac{1}{2} (2+(-1))^n = \frac{1}{2} (5^n - 1)$ $\therefore B_n = 2^n \cdot \frac{1}{2} \begin{pmatrix} 5^n + 1 & 5^n - 1 \\ 5^n - 1 & 5^n + 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 10^n + 2^n & 10^n - 2^n \\ 10^n - 2^n & 10^n + 2^n \end{pmatrix}$ (c) (i) $\det(B_n) = \frac{1}{4} [(10^n + 2^n)^2 - (10^n - 2^n)^2]$ $= \frac{1}{4} [(10^n + 2^n + 10^n - 2^n)(10^n + 2^n - 10^n + 2^n)] = \frac{1}{4} (2 \cdot 10^n)(2 \cdot 2^n)$ $= 10^n \cdot 2^n = 20^n$ (ii) $\begin{pmatrix} x_{P'} \\ y_{P'} \end{pmatrix} = B_n \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 10^n + 2^n \\ 10^n - 2^n \end{pmatrix}, \begin{pmatrix} x_{Q'} \\ y_{Q'} \end{pmatrix} = B_n \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 10^n - 2^n \\ 10^n + 2^n \end{pmatrix}$ Area of $\triangle OP'Q' = \frac{1}{2} x_{P'} y_{Q'} - x_{Q'} y_{P'} = \frac{1}{2} \det(B_n) = \frac{1}{2} (20^n)$ Area of $\triangle OPQ = \frac{1}{2} 1 \cdot 1 - 0 \cdot 0 = \frac{1}{2}$ $\therefore$ Area of $\triangle OP'Q' = 20^n \times$ Area of $\triangle OPQ$	1M 1A 1M 1A 1M 1A	

Solution	Marks	
12. By the Binomial Theorem: $a_n = \sum_{r=0}^n C_r^n (\sqrt{3})^r = (1 + \sqrt{3})^n$ $b_n = \sum_{r=0}^n C_r^n (-\sqrt{3})^r = (1 - \sqrt{3})^n$ $\vec{u}_n \cdot \vec{v}_n = (a_n \mathbf{i} + 2^n \mathbf{j}) \cdot (b_n \mathbf{i} - \mathbf{j}) = a_n b_n - 2^n$ $a_n b_n = (1 + \sqrt{3})^n (1 - \sqrt{3})^n = [(1 + \sqrt{3})(1 - \sqrt{3})]^n = (1 - 3)^n = (-2)^n$ Since n is a positive even integer, $(-2)^n = 2^n$. $\therefore \vec{u}_n \cdot \vec{v}_n = 2^n - 2^n = 0$ Geometric relationship: $\vec{u}_n$ is perpendicular (or orthogonal) to $\vec{v}_n$ (i.e. $\vec{u}_n \perp \vec{v}_n$).	1M 1M 1M 1A 1A	